CY4001: Industrial Chemistry

Unit 2: Industrial Safety & Hazards Management

Chemistry of toxicity/poisoning

Toxicology in the Movies: Fact Check Time!

Do poisons cause people to vomit blood like in movies?

Coloured smokes?

Foaming?

Instant Collapse?

Eye colour changes?

"The Antidote" That Cures Everything Instantly?

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Industrial Toxicology

Industrial toxicology is the study of the harmful effects on humans by chemicals used in the workplace, the products produced by companies, and the waste created in manufacturing.

A poison is any substance that causes harm by blocking, disrupting, or overwhelming the body's chemistry.

- It blocks an essential reaction.
- It mimics the wrong molecule.
- It overstimulates or shuts down a pathway.
- It binds to something it shouldn't.
- It disrupts cell energy, oxygen, or signaling.

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Industrial Toxicology

Industrial chemicals that cause the most harm to the body are classified as **irritants, asphyxiants,** and **poisons**

Irritants

- Irritants are substances that cause inflammation, rashes, or corrosion of skin, lungs etc. They can also cause pain, swelling, mucus secretion, and muscle constriction.

  E.g.: Gases such as chlorine, ammonia, Sulfur dioxide and formaldehyde

- Particulate irritants are irritants that are composed of minute bits of material produced within the work environment.

E.g. Coal miners are exposed to silica, which can result in pneumoconiosis, silicosis (black lung disease), and asbestosis.

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Industrial Toxicology

How Do Irritants Work? (Chemistry Behind the Annoyance)

- Breaking or reacting with proteins: Many irritants react with amino acids at the surface of cells, acid–base reactions, oxidation / reduction reactions

- Disrupting lipid membranes: They disturb the protective barrier

Industrial Safety

- ❑ Ventilation
- ❑ PPE (gloves, masks, goggles)
- ❑ Exposure limits (TLV, OSHA guidelines)
- ❑ Proper storage
- ❑ Training and first aid

Irritants may seem mild, but **chronic exposure = chronic problems**.

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Industrial Toxicology

Systemic Poisons: "Whole-Body Trouble Makers"

Systemic poisons act on the whole body, interfering with fundamental biochemical pathways.

- ❑ Cyanide → blocks cellular respiration throughout the body
- ❑ Hydrogen sulfide (H_2S) → systemic cellular toxin
- ❑ Carbon monoxide (CO) → affects all oxygen-dependent tissues
- ❑ Arsenic → disrupts multiple systems (GI, nerves, blood)

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Industrial Toxicology

Organ Poisons: "Targeted Assassins"

Organ poisons target **specific organs**, depending on their metabolism, accumulation, or binding affinity (e.g., liver toxins, kidney toxins, neurotoxins).

They attack the organs where they accumulate, activate, or bind.

Neurotoxins (Brain & Nerves)

- Lead
- Mercury
- Organophosphate pesticides

Hepatotoxins (Liver)

- Industrial solvents (CCl_4)
- Some alcohol metabolites

Respiratory Poisons (Lungs)

- Chlorine
- Ammonia
- Nitrogen oxides

Nephrotoxins (Kidneys)

- Cadmium
- Chromium
- Some heavy metals

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Industrial Toxicology

Asphyxiants: "The Oxygen Thieves" of Toxicology

Asphyxiants are substances that cause harm by interfering with your ability to breathe or use oxygen.

- Simple Asphyxiants: These chemicals displace oxygen within the lungs, thus inhibiting the amount of oxygen being transported throughout the body to nourish cells.
- **Example:** Nitrogen and helium, used to flush vats and tanks before routine maintenance, are examples of simple asphyxiants.

- Chemical Asphyxiants: They create harm when inhaled because they create a chemical reaction in the body.
  Example 1: Carbon monoxide (CO), for example, robs the body of oxygen by competitively binding to hemoglobin in the blood and preventing oxygen being transported to tissues of the body.
  Example 2: Hydrogen sulfide (H_2S), on the other hand, paralyzes the muscles in the lung and throat and prevents oxygen from entering the body.

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Industrial Toxicology

Asphyxiants: "The Oxygen Thieves" of Toxicology

Safety Takeaway

Industries prevent asphyxiant exposure with,

- ❑ Oxygen monitoring sensors
- ❑ Ventilation
- ❑ Confined-space protocols
- ❑ Respirators
- ❑ Training

In toxicology, the first rule is simple but powerful:

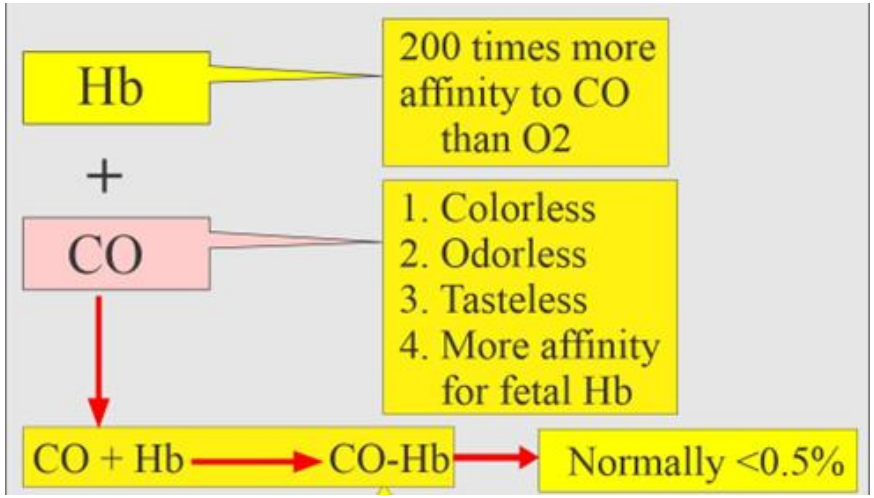
"Always trust the oxygen meter, never your nose."

CO poisoning: The Invisible Trickster

Carbon monoxide has no smell, no color, no taste.

The Invisible Trickster

- Carbon monoxide binds very strongly to hemoglobin.
- Once bound, it blocks oxygen transport
- Blood looks normal, Lungs are working
- Oxygen is present
- But tissues are quietly starving.



Because CO is colorless, odorless, and non-irritating, exposure often goes unnoticed, leading to symptoms like headache, dizziness, confusion, and at high levels, rapid unconsciousness.

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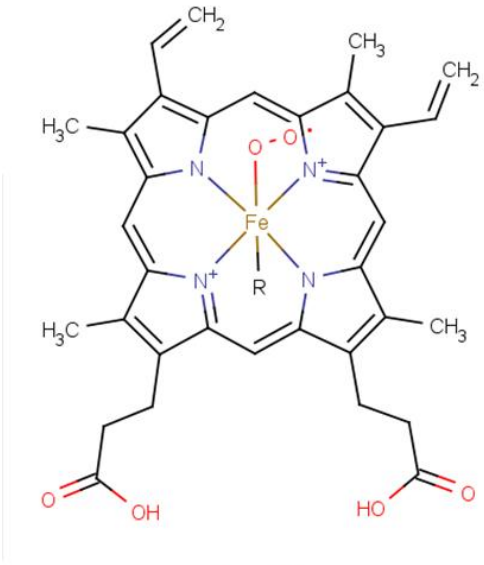
CO poisoning: The Invisible Trickster

- As soon as it enters the body, Carbon monoxide combines with hemoglobin to form carboxyhemoglobin (HbCO) preventing the blood from carrying oxygen.

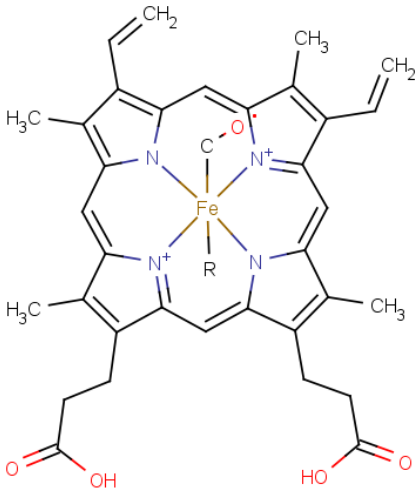
- Hemoglobin has 200 time more affinity to CO than O_2

- In addition to that many other hemoproteins such as myoglobin, Cytochrome P450, and mitochondrial cytochrome oxidase are also affected.



Oxyhemoglobin

CO →



Carboxyhemoglobin

Cyanide poisoning: ("When Oxygen Is There… but Useless")

Cyanide is dangerous because it **blocks cellular respiration**.

Histotoxic hypoxia

- Oxygen reaches the cells
- But cells cannot use oxygen
- Energy production (ATP) stops
- Cells "suffocate" even though oxygen is present

- CO blocks oxygen delivery
- Cyanide blocks oxygen utilization

- Cyanide does not block oxygen from entering the body; it blocks cells from using oxygen by inhibiting key mitochondrial enzymes (especially cytochrome c oxidase), causing histotoxic hypoxia.
- Oxygen levels in blood may be normal, but energy production (ATP) stops, leading to rapid failure of high-energy organs such as the brain and heart.
- Onset is rapid and severe, with symptoms like headache, confusion, rapid breathing, and sudden collapse in high exposure, making cyanide one of the most dangerous industrial and combustion-related toxins.

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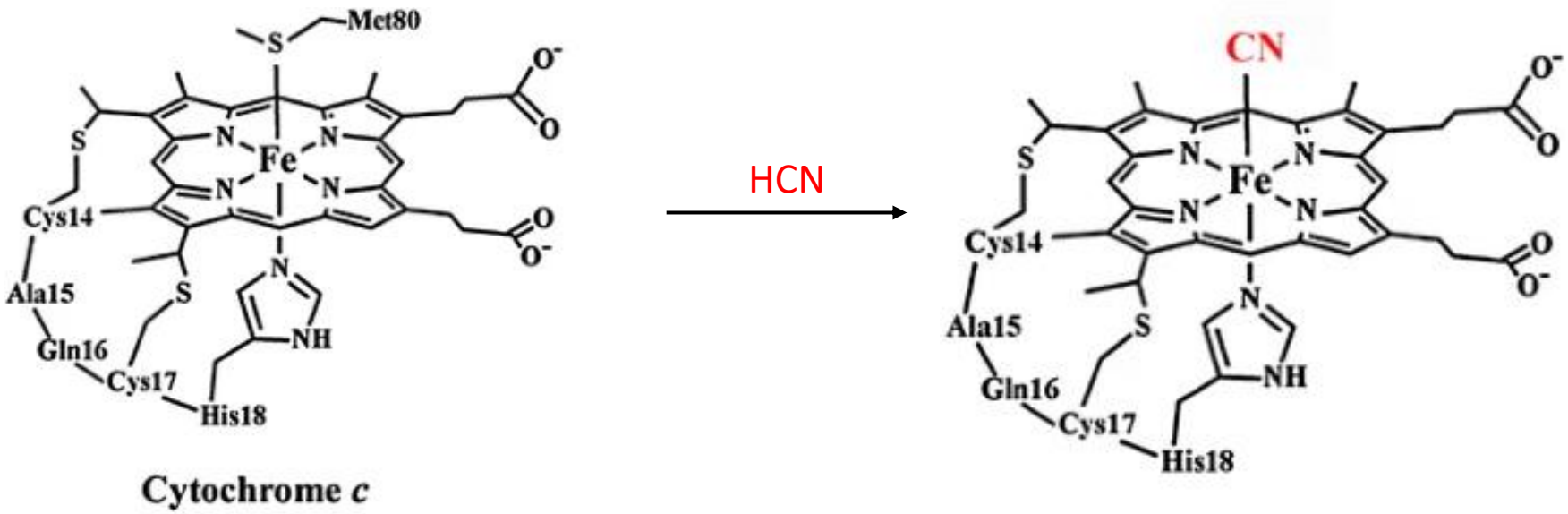
Cyanide poisoning

- Cyanide's main target is **cytochrome c oxidase** in mitochondria
- This enzyme contains iron (Fe^{3+}) at the active site
- Cyanide binds tightly to this enzyme and shuts down cellular respiration



Cytochrome *c*

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H_2S poisoning: ("The Rotten Egg Assassin")

- Hydrogen sulfide is a toxic gas with a characteristic rotten-egg smell, but at higher levels it causes olfactory fatigue, making the gas undetectable by smell and therefore extremely dangerous.

- H_2S inhibits cellular respiration by blocking key enzymes in the mitochondria, preventing cells from using oxygen to produce energy, leading to histotoxic hypoxia.

- Exposure primarily affects the nervous and respiratory systems, and high concentrations can cause sudden collapse, especially in confined industrial spaces such as sewage systems and refineries.

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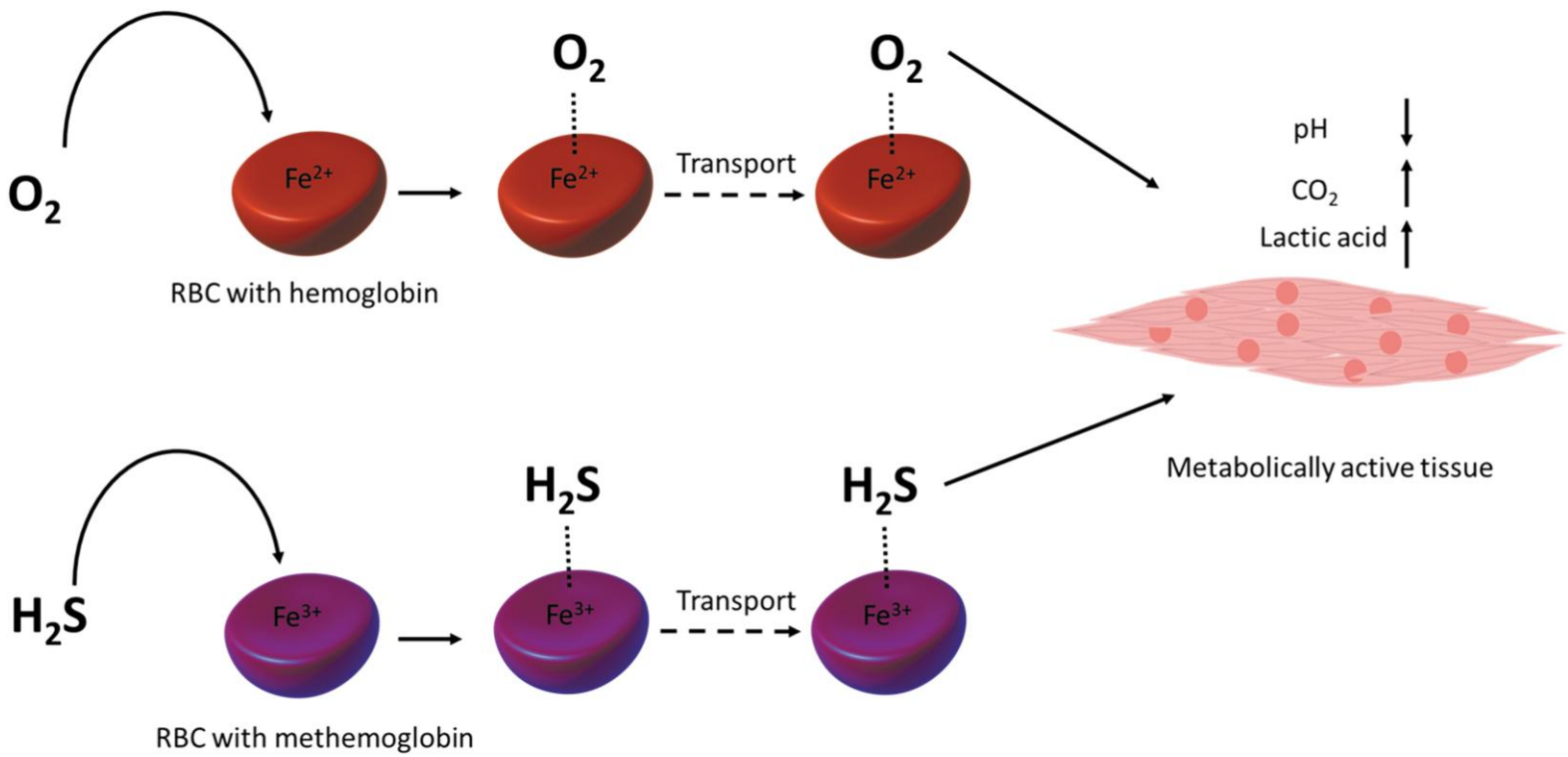
H_2S poisoning: ("The Rotten Egg Assassin")

Hydrogen sulfide inhibits mitochondrial cytochrome C oxidase by making a complex bond to the ferric moiety of the protein, thereby arresting aerobic metabolism and blocking ATP synthesis. This mechanism of toxicity is similar to hydrogen cyanide.



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Metal toxicity - When Useful Metals Turn Harmful

Metals are essential in small amounts, but dangerous when chemistry loses control.

What is Metal Toxicity?

Metal toxicity occurs when metals accumulate in the body and interfere with normal biochemical processes, leading to damage of cells, enzymes, and organs.

- ❑ Metals do not break down easily
- ❑ They persist and accumulate
- ❑ Their effects are often chronic and long-lasting

The Chemistry Behind

- ➢ Binding to enzymes and proteins
- ➢ Displacing essential metals
- ➢ Generating oxidative stress
- ➢ Accumulating in specific organs

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Lead poisoning: "The Silent Neuro-Villain"

Lead poisoning is a classic example of chronic metal toxicity: slow, sneaky, and widespread.

How Lead Behaves in the Body

- Mimics essential metals: Lead pretends to be Calcium (Ca^{2+}), Zinc (Zn^{2+}), Iron (Fe^{2+})
- Because of this mimicry, it enters bones (replaces calcium), neurons (blocks neurotransmission) and enzymes (blocks heme synthesis)
- Inactivates enzymes and shuts down key biochemical pathways, especially heme synthesis, causing anemia
- Accumulates for years - Lead is the guest who never leaves the house.

Arsenic poisoning

WEIRD HISTORY

The Most Weaponized Poison In World History: The Deadly Trail Of Arsenic Through The Ages

Arsenic doesn’t cause instant collapse: it causes slow chaos in the body’s chemistry.

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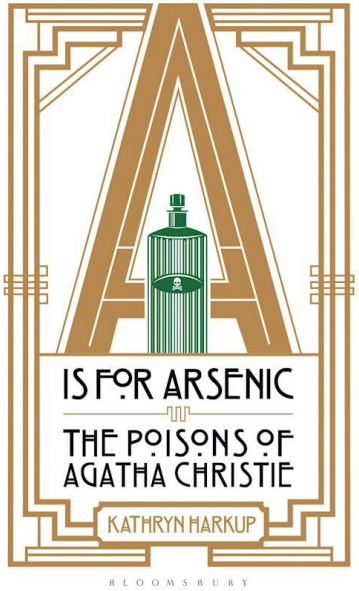
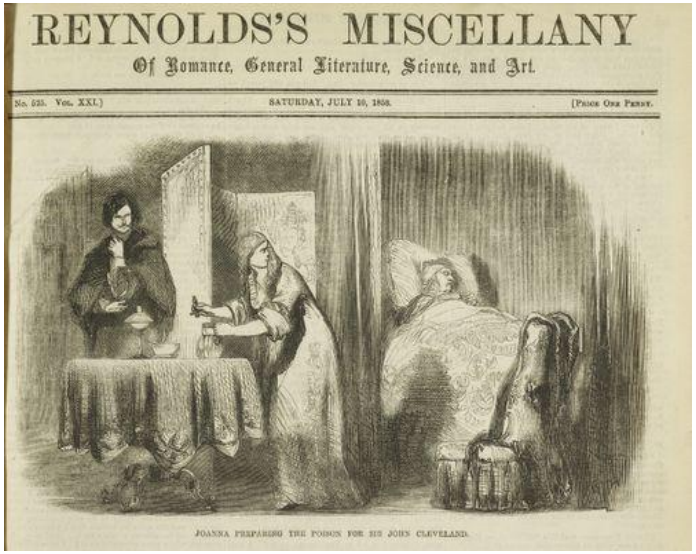
Arsenic poisoning

- Arsenic (more properly arsenic trioxide) is a highly poisonous white powder. It was very cheap, a by-product of metal industries.

- Arsenic was a convenient secret weapon for poisoners as the main symptoms of arsenic poisoning were vomiting and diarrhoea.

Arsenicosis is a medical condition that occurs due to elevated levels of arsenic in the body. Arsenic acts by changing the functioning of around 200 enzymes.





Arsenic poisoning

Arsenic mainly exists in two toxic forms: **Arsenite (As^{3+})** and **Arsenate (As^{5+})**. Both disrupt metabolism, but in different ways.

Arsenite (As^{3+}): Binds to proteins

- Likes attaching to sulfhydryl (–SH) groups in enzymes
- Shuts down essential metabolic pathways
- Stops ATP production
- Affects skin, nerves, and the GI tract

Arsenate (As^{5+}): Mimics phosphate

- Enters biochemical pathways by pretending to be phosphate
- Disrupts energy production (ATP)
- Interrupts with DNA and metabolic reactions

- Arsenic disrupts essential enzymes and ATP production, either by binding to sulfhydryl groups (As^{3+}) or by mimicking phosphate (As^{5+}).
- Chronic exposure causes skin changes, neuropathy, and cancer, while acute exposure leads to severe GI distress and cardiovascular collapse.

Mercury Poisoning

Mercury is unique because it exists in different chemical forms, and each form behaves differently in the body.

Chemistry decides toxicity

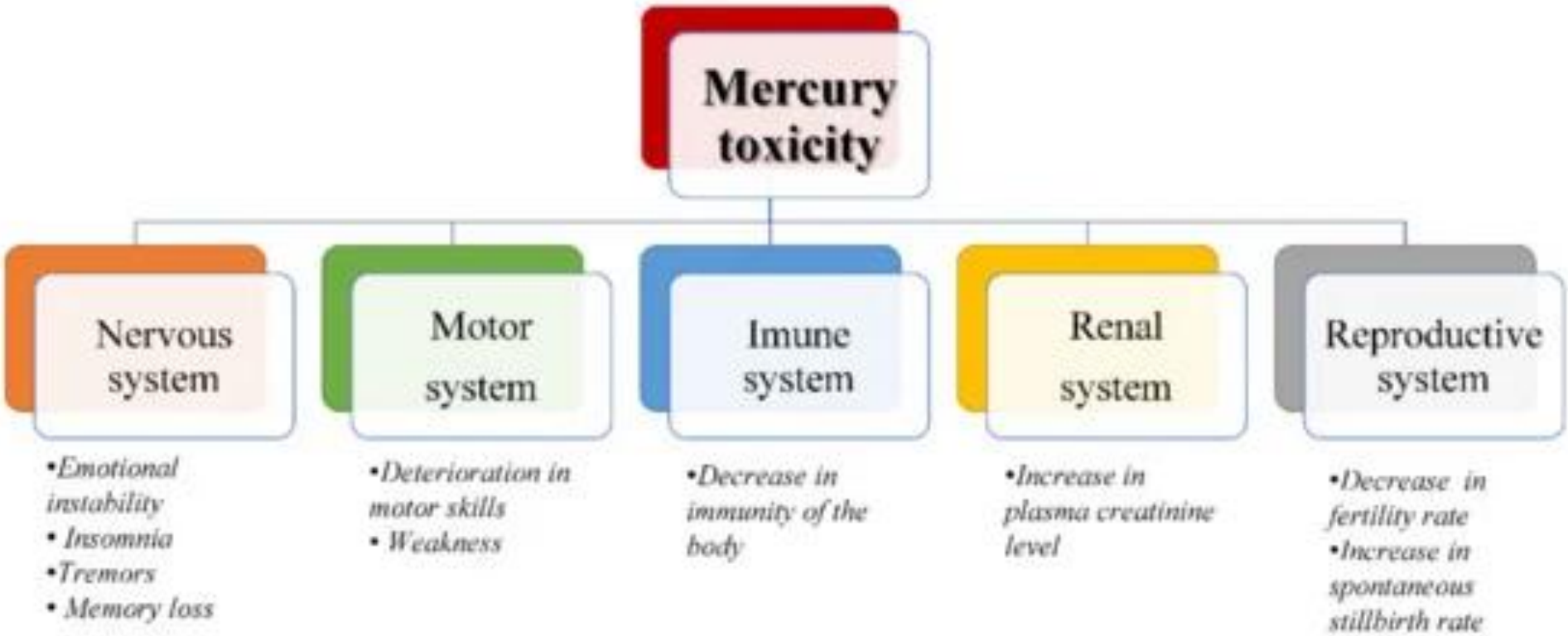
Mercury toxicity depends strongly on its chemical form

- Elemental mercury (Hg^0) - vapour form
- Inorganic mercury (Hg^{2+} salts)
- Organic mercury (methylmercury) - the most dangerous form

How Mercury Causes Toxicity

- Mercury binds strongly to sulfhydryl groups, disrupting enzymes and proteins, with severe effects on the nervous system.
- Organic mercury (methylmercury) is the most toxic form, easily crossing the blood-brain barrier and placenta.
- Chronic exposure leads to neurological and kidney damage, making mercury a serious industrial and environmental hazard.

Mercury Poisoning



https://doi.org/10.3390/ijms23041992

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Chromium Poisoning: "Same Element, Very Different Chemistry"

In chromium, the oxidation state decides whether it helps life or harms it.

Chromium (III) – Cr^{3+}

- ❑ Essential trace element
- ❑ Involved in glucose and lipid metabolism
- ❑ Required only in very small amounts
- ❑ Generally low toxicity

Chromium(VI) – Cr^{6+}

- ❑ Highly toxic and carcinogenic
- ❑ Strong oxidizing agent
- ❑ Readily enters cells
- ❑ Responsible for most chromium poisoning cases

How Cr Causes Toxicity

- ➢ Cr^{6+} enters cells easily and generates oxidative stress, damaging DNA, proteins, and membranes.
- ➢ Chronic industrial exposure leads to respiratory disease and cancer, making strict regulation and protection essential.

Antidotes: "The Counterattack to Poisons"

An antidote is a substance that reduces or reverses the harmful effects of a poison by blocking it, neutralizing it, or helping the body get rid of it.

- Antidotes counteract poisons by blocking their action, neutralizing them, or enhancing their elimination.
- Most antidotes work through chemical specificity, meaning they target a particular poison or mechanism.
- Antidotes must be professionally administered, as they can affect normal biochemical systems if misused.

Antidotes: "The Counterattack to Poisons"

Cyanide poisoning	Hydroxocobalamin: It works by binding directly with cyanide to form cyanocobalamin (a stable form of Vitamin B12), which is then safely excreted by the kidneys
CO poisoning	Administration of 100 percent oxygen through a nonrebreathing mask until COHb levels fall below 5 percent
H_2S poisoning	No proven antidote for hydrogen sulfide poisoning
Metal poisoning	Chelation theraphy

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Chelation therapy: "Removing Toxic Metals with Chemistry"

Chelation therapy is the process of using special molecules to grab toxic metals and help the body remove them safely.

What Is Chelation?

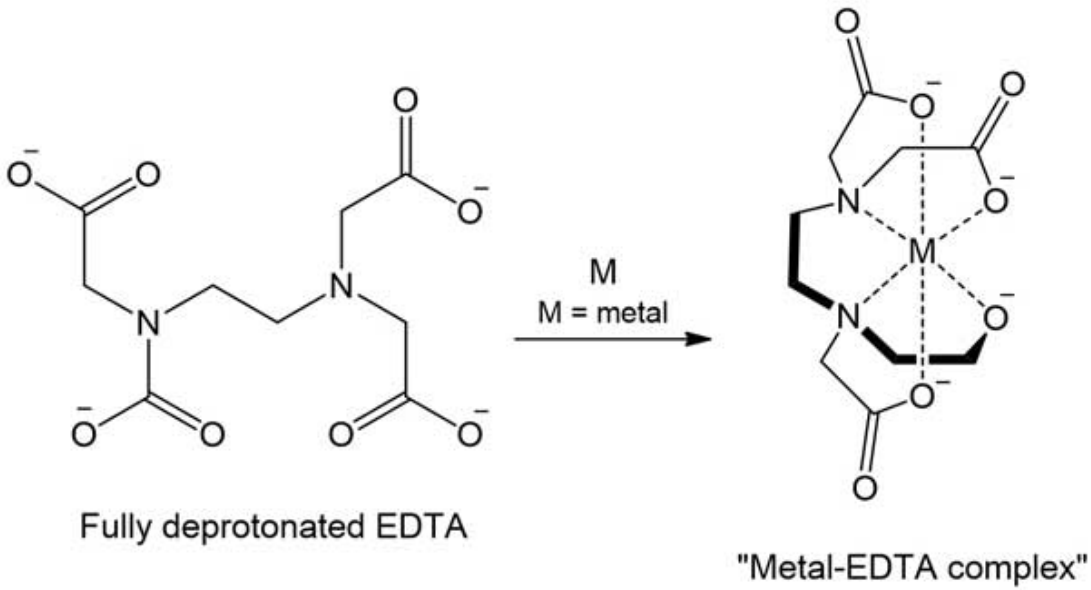
Chelators are molecules that have multiple binding sites (like claws).
They can hold onto metal ions tightly and form a stable, ring-like complex called a *chelate*.

How Chelation Therapy Works

- A chelating agent binds the toxic metal strongly
- This forms a metal-chelator complex
- The complex is excreted by the body

Chelation therapy

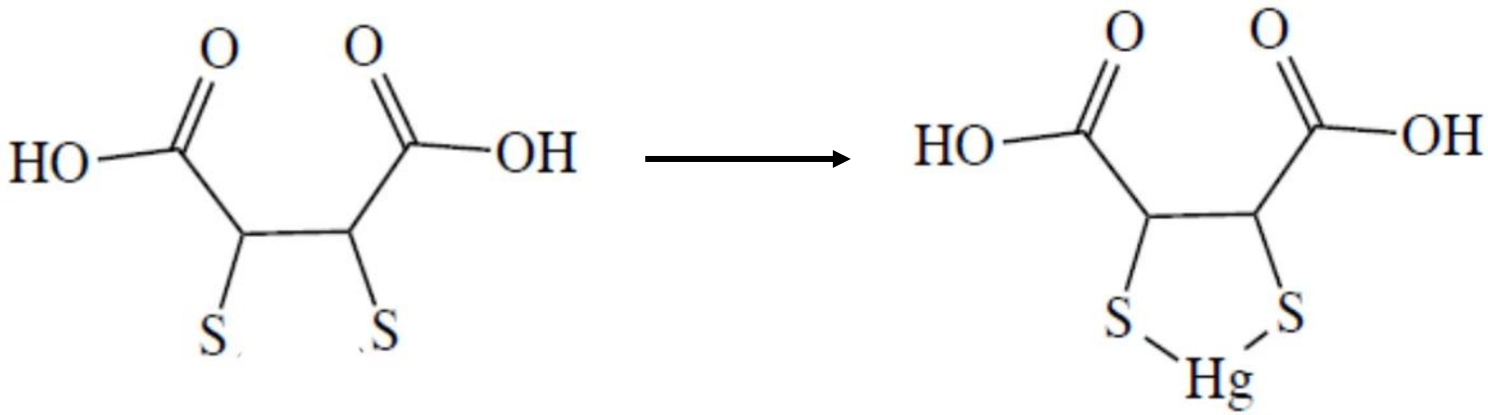
- Chelation is the formation of multiple coordination bonds between organic molecules and a transition metal ion leading to sequestration of the metal.



EDTA is a chelating agent that binds to calcium, lead, iron, copper, and other metal ions forming soluble complexes and facilitating their urinary excretion.

Treatment for Mercury Poisoning

- Dimercaptosuccinic acid (DMSA) is a chelating agent used to treat heavy metal poisoning, such as mercury. It's taken orally and is considered safe and effective.
- DaMSA binds to heavy metals in the body, such as lead and mercury and remove the metals from soft tissues, such as the brain
- DMSA increases the amount of lead excreted in urine, which can reduce blood lead levels

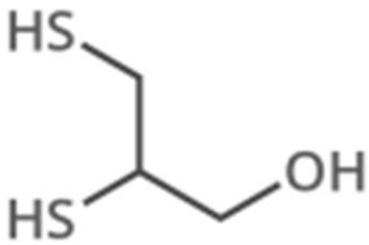


DMSA

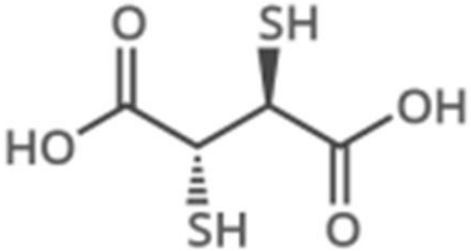
DMSA –chelated mercury

Treatment for Arsenic Poisoning

Chelating agents such as Dimercaprol and Dimercaptosuccinic acid bind the arsenic ions and prevent them from inhibiting enzymes.



DIMERCAPROL



DIMERCAPTOSUCCINIC ACID

If poison reaches it's expiry date

would it be more poisonous or less poisonous?

"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

– Richard Feynman

Thank you